

Visualizing the Carbon Intensity of Machine Learning Inference for Image Analysis on TensorFlow Hub

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ABSTRACT

The increasing performance of machine learning (ML) models necessitates greater computing resources, contributing to rising carbon intensity in ML computing and raising concerns about computational equity. Previous studies focused on developing tools that enable model developers to view the carbon intensity of the ML models in the training process. Still, little is known about how to support ML developers in online communities to explore the carbon intensity of ML models during inference. We developed MIEV, a model inference emission visualizer, that supports ML developers on TensorFlow Hub to explore the carbon intensity of image domain models during the model **Inference** phase. We also provide insights into designing technologies that promote collaborative work among ML developers to drive sustainable AI development processes. To the best of our knowledge, this is the first attempt to interactively visualize the carbon intensity of ML models in online communities during the **Inference** phase.

CCS CONCEPTS

• **Human-centered computing** → **Visualization systems and tools**.

KEYWORDS

TensorFlow Hub, online communities, inference, carbon intensity

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1 INTRODUCTION

The continuous progress of ML models has substantially improved their modelling performance, largely attributed to the utilization of large architectures that effectively handle complex datasets [14]. Nevertheless, the usage of such large models requires substantial computational resources, presenting an environmental concern as datasets grow in complexity. An illustrative instance is the exponential growth in the number of parameters within ML models, which surged approximately 300,000-fold between 2012 and 2019, resulting in a significant surge in carbon emissions. Furthermore, the carbon intensity of a training session of the popular GPT-3 LLM is similar to the carbon intensity generated when flying from San Francisco to New York City and back three times ($\sim 1.2t$ CO₂ equivalent) [36]. The growth of carbon emission leads to the negative effects of climate change which results in increased environmental and financial costs impacting the world's most marginalized countries. Hope stated that social cost created by carbon emissions such as climate changes and uncertain abatement costs is expected to rise about 3.2% per year continuously [21]. Considering the growing trend of training ML models on cloud computing instances, it becomes essential to address the emissions generated by these workloads to effectively reduce carbon intensity within the field.

To reduce carbon intensity generated during the **Training** phase, researchers proposed visualization tools enabling ML developers to explore the carbon intensity [14, 27, 30, 35, 36]. For example, Dodge et al. provided a framework for measuring carbon intensity when ML models such as BERT [13], and DenseNet [22] are trained [14]. Also, previous studies created tools for ML developers to view the carbon intensity of ML models during the training process [2, 5, 20, 27, 28, 32, 38]. For instance, Shaikh et al. developed EnergyVis, an interactive tool for ML developers to track the energy consumption of ML models and explore alternative deployment locations [38]. Despite the fact that the **Inference** phase accounts for 70% of the whole ML model development process (i.e., **Data Processing**, **Experimentation**, **Training**, and **Inference**), previous studies focused mostly on measuring and visualizing the carbon intensity in the **Training** phase [46]. In this respect, Verdecchia et al. reported, it is imperative to devise methodologies that enable model developers to incorporate considerations of carbon emissions throughout the entirety of ML model development, extending beyond solely the **Training** phase [44].

In addition, online communities, such as TensorFlow Hub (TF Hub) [42], PyTorch Hub [37], and Hugging Face [15] enable ML developers to share pre-trained ML models. These models then undergo the Inference phase as part of the model development process. With the proliferation of ML models being shared online, it is foreseeable that a vast number of models will go through the Inference phase, requiring substantial computational power and leading to a significant carbon emission footprint [4, 44]. Thus, model developers need to view the carbon intensity of ML models during the **Inference** phase when using these models. Nonetheless, almost no studies proposed visualization tools that enable ML developers to explore the carbon intensity of models in online communities during the **Inference** phase.

To address this gap, we developed MIEV, a model inference emission visualizer that enables ML developers to explore the carbon intensity of ML models in online communities during the model **Inference** phase. In this study, we focused on the image classification models in the image domain of TF Hub, specifically on the model trained using the ImageNet-21k dataset. Our study makes the following contributions to the CSCW community: (1) We develop a model inference emission visualizer (MIEV) that demonstrates exploring the carbon intensity for ML models on TF Hub during the **Inference** phase; (2) We present use cases of ML developers considering the carbon intensity in the model **Inference** phase; (3) We provide design insights toward technical development that supports collaborative work among ML developers to lead sustainable AI development processes.

2 RELATED WORK

In this section, we summarize prior studies which focused on understanding online communities of ML developers, developing visualization tools for ML developers and tools for tracking energy consumption of ML models during the **Training** phase. We also present the limitations of previous studies and the importance for ML developers to consider carbon intensity when developing models in online communities.

In prior work, researchers explored the online communities of ML developers such as Stack Overflow [1, 3, 8, 19, 23, 24, 49, 50], TF Hub [6, 48], and GitHub [12, 23]. Four studies identified the problems and challenges that ML developers face [1, 8, 24, 49], while others reported the trends of ML on Stack Overflow [3, 19, 50]. For example, Zhang et al. reported the challenges ML developers face when building deep learning applications by analyzing the postings on Stack Overflow [49]. On the other hand, researchers investigated the community of ML developers on TF Hub [6, 48], while others explored the communities, such as GitHub [24], and personal webpages [12]. For instance, Yoo et al. reported model publishers and the models on TF Hub to explore how ML developers share their models on TF Hub [48].

Researchers proposed visualization tools that support ML developers during the ML model creation process [7, 9, 18, 25, 26, 29, 31, 34, 41, 43, 45, 47, 51]. Several studies proposed visualization tools to help ML developers prepare data for model training [29, 41, 51]. For instance, Sun et al. created a visualization tool designed to assist ML developers in understanding the behavior of a classifier during

the labeling task [41]. Also, there are other types of tools that visualize data prediction results after training models or visualize data architecture before training models. While researchers developed visualization tools to help ML developers understand how ML models make predictions of ML models that are trained by using graph data [25, 26, 45, 47], tabular data [7, 9, 34], and wave [18], and text data [31], other researchers proposed visualization tools that help ML developers by visualizing the data flow before model training (e.g., a GUI framework allowing ML developers to visualize the data flow in your architecture before model training anomalies) [43].

Several studies were proposed to help ML developers measure the carbon intensity during the ML model **Training** phase, by developing library packages [2, 5, 20, 32], and web applications [27, 28, 38]. For example, Shaikh et al. proposed EnergyVis, and web application for ML developers to track energy consumption in training ML models [38]. It supports model developers to interactively compare the energy consumption of ML models by taking into account hardware specifications and the location of deployment location.

Based on the aforementioned studies, little is known about the visualization tool that allows ML developers to explore the carbon intensity during the model **Inference** phase. It is important for ML developers to become aware of the carbon intensity of the ML models' inference processes since the inference process accounts for 70% of the carbon intensity of the whole ML model development process [46]. Also, prior studies did not measure the carbon intensity of ML models from online communities during the **Inference** phase. As Yoo et al. discussed, it is essential to help ML developers interact with each other in online communities such as TF Hub, to build trustworthy and responsible ML developer communities [48]. Therefore, our aim is to develop an interactive visualization tool that supports ML developers to explore the carbon intensity of ML models in TF Hub during **Inference** phase, ultimately fostering trustworthiness through improving explainability [39].

3 MIEV DESIGN

MIEV is an interactive visualization tool that allows ML developers to compare the carbon intensity of models on TF Hub produced during the **Inference** phase. MIEV consists of a frontend and backend implemented with Flutter [17] and Flask [16] respectively. The overview of MIEV user interface is illustrated in Figure 1.

3.1 Backend Implementation

The backend of MIEV loads the models selected in the frontend from the TF Hub, makes inferences on the uploaded data, and measures the carbon intensity. The backend of the tool provides an API that performs the following tasks: display a list of models, display the inference results, and measure the carbon intensity based on the model and image selected by the ML developer. Get the list of available models from TF Hub and display them. In this study, the model list is limited to the models trained on the ImageNet-21k dataset. The carbon intensity of the model's **Inference** phase was calculated using CodeCarbon Library [10].

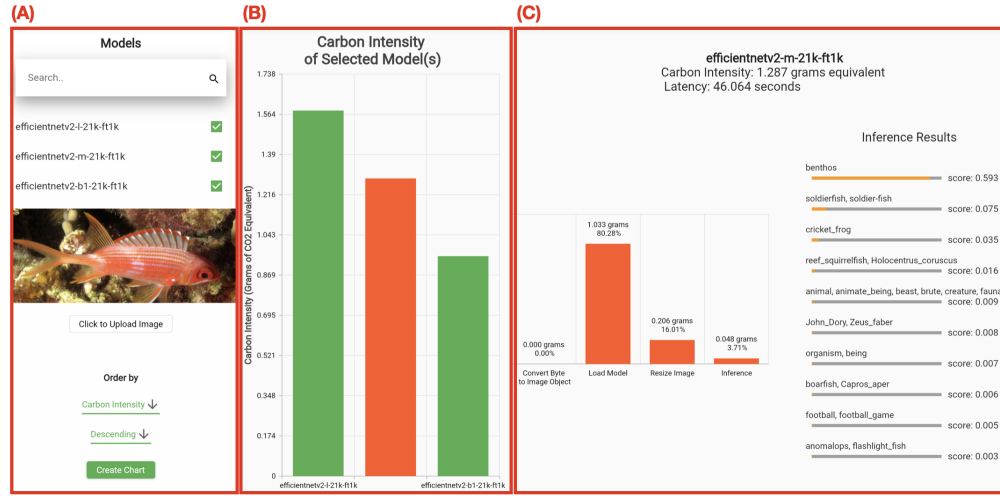


Figure 1: The user interface of MIEV. (A) ML developers select models from TF Hub and upload the image for inferring class and probability. (B) Then, MIEV enables ML developers to compare the carbon intensity generated during the Inference phase. (C) Also, it allows ML developers to view the carbon intensity of each step of the Inference phase.

3.2 User Interfaces

MIEV enables ML developers to select models published on TF Hub and upload an image to infer (see Figure 1-(A)), explore the carbon intensity in the ML model **Inference** phase (see Figure 1-(B)), and view the ratio of the carbon intensity on each inference step and inference results (see Figure 1-(C)). The rules of data visualization, such as overview, zoom and filter, and detail-on-demand were applied [40].

3.2.1 Selecting Models Published on TF Hub. MIEV allows ML developers to select models from TF Hub to measure the carbon intensity of their inference (see Figure 1-(A)). In addition, it enables users to search for the models by entering search terms. Following that, MIEV enables users to upload an image for model inference. In addition, ML developers are allowed to sort the chart in Figure 1-(B) either by carbon intensity or alphabetically based on the name of the selected models. After selecting the model and uploading the image, ML developers are able to click the "Create Chart" button to see the carbon intensity generated by the model during the **Inference** phase.

3.2.2 Visualizing the Carbon Intensity of Models. After clicking the "Create Chart" button, MIEV allows ML developers to explore and compare the carbon intensity that selected models produce when inferring the uploaded image (see Figure 1-(B)). On the x-axis of the chart, ML developers are able to see the names of the selected models. The y-axis shows the carbon intensity (CO2 equivalent) in grams, generated by the **Inference** phase of the selected models. The carbon intensity is the sum of the four steps in the model's inference process: (1) converting binary to the image object, (2) loading the selected TF Hub model, (3) resizing the image to match the input size, and (4) returning the inference result.

3.2.3 Visualizing Details. When ML developers select one of the bars visualizing the carbon intensity of models (see Figure 1-(B)),

MIEV enables them to view the details (see Figure 1-(C)) for the selected model. When ML developers click one of the bars in Figure 1-(B), the bar is emphasized by displaying it in orange color, allowing ML developers to view details of the **Inference** phase. It consists of two sub-charts. The first chart shows the percentage of carbon intensity that occurs at each of the four stages of the model's **Inference** phase (see Figure 1-(C) left). The second chart allows ML developers to view the inference result. The results contain the latency, the name, and the score of each label, sorted in descending order by score (see Figure 1-(C) right).

4 USAGE SCENARIOS

Our vision for MIEV involves engaging user communities in computational tasks, especially those with limited access to environmental impact information or visualization. To illustrate this, we present two usage scenarios in a context where carbon taxes are imposed on ML model activities and communities. We suggest two personas, a TF Hub user and an ML developer. Both have a similar goal: finding an ML model of low carbon intensity in the phase of inference.

Comparing carbon intensity when inferring models on TF Hub: Olivia, a graduate student, needs to deploy an image classification model for her research project. Olivia selects the three models as candidates: (a) bit/m-r50x1/imagenet21k_classification, (b) imagenet/efficientnet_v2_imagenet21k_s/classification, and (c) bit/m-r101x3/imagenet21k_classification. Despite limited knowledge of the model's structure, Olivia aims to consider both the classification accuracy and the carbon intensity generated during the **Inference** phase when selecting a candidate model. Unable to find the carbon intensity information for the models on TF Hub, Olivia visits the MIEV website to compare the carbon intensity of each model during inference. She selected the three candidate models from the website and uploaded images to test the classification results. And she compared the carbon intensity generated when

the three models classified images in exploring the carbon intensity chart (see Figure 1-(B)). Olivia also compared the classification performance and latency of the models by looking at the results inferred by each model in the viewing details (see Figure 1-(C)). She found that model (a) had slightly higher classification accuracy than other models, but that model (c) had lower carbon intensity. Olivia finally chose model (c) to deploy on her project in consideration of the carbon intensity.

Comparing the carbon intensity generated at each step in the model Inference phase: Bruno is an ML developer working on developing image classification models. The company he works for develops ML models and shares the model on TF Hub. Due to government regulations and the corporation's Sustainable Development Goals, the company is trying to reduce the amount of carbon intensity generated by image classification models. Bruno launches MIEV, selects models (a), (b), and (c) that he uploaded to TF Hub, and uploads example images for classification. He then compares the carbon intensity of the three image classification models, considering the carbon emissions generated during the **Inference** phase of each model. As a result, Bruno found that model (b) had the highest carbon intensity when inferring the image. Bruno clicked model (b) on exploring the carbon intensity chart (see Figure 1-(B)) to find out which step has a high carbon intensity. He found that when model (b) received the resized image and returned the inference result, the carbon intensity was higher than that of other models. Hence, Bruno planned to decrease the carbon intensity produced by the model (b) when it returns inference results as a follow-up research goal.

5 DISCUSSION

We discuss how to design MIEV to better assist ML developers in developing ML models while considering carbon intensity. First, while MIEV facilitates the development of ML models with environmental considerations by comparing the carbon intensity associated with models from TF Hub, it is still unclear how ML developers in the online community can effectively integrate carbon intensity into their development process. Recent research highlights that encouraging collaboration among ML developers can positively impact the level of engagement within online communities, including platforms like GitHub. Marlow et al. found that developers engaged in various activities online, including showcasing their interests by following features, discovering new projects, and seeking new information from followers [33]. Moreover, Dabbish et al. demonstrated that visible actions, such as following and liking, are able to foster a sense of appreciation and motivation among developers [11]. Thus, our vision for an integrated MIEV within an ML model-sharing community is to enable ML developers to contribute not only to model performance but also to the reduction of carbon intensity. By fostering collaboration and providing ML developers with tools for considering environmental impacts, MIEV can empower developers to collectively work towards more sustainable and efficient ML model development.

Second, it is still questionable how to reduce the carbon intensity in this process. Previous studies proposed methods mitigating the carbon intensity generated during the model training process (e.g., measuring carbon intensity [14], visualizing carbon intensity [38]).

By providing ML developers with alternative deployment options that take carbon intensity into account, we can potentially motivate them to consider the environmental impact of their models throughout the development process. Offering these options within MIEV would encourage developers to make more conscious choices and prioritize sustainability when deploying their ML models.

6 LIMITATIONS AND FUTURE WORK

This study has several limitations. In our study, we focused on models trained with the ImageNet-21k dataset specifically within the image classification category available in TF Hub. However, as Yoo et al. reported that TF Hub encompasses four domains, namely image, text, audio, and video, each with various categories such as classification and embedding [48]. Therefore, further exploration and investigation are needed in these domains to better understand model performance and carbon intensity. Additionally, evaluating MIEV with real users would enhance its usability. User-centered evaluation can provide insights into the practicality and effectiveness of MIEV in real-world scenarios, potentially leading to improvements and refinements in its design and functionality.

Future work remains to extend the functionality of MIEV beyond image classification in the image domain to compare the carbon intensity of models in various categories, domains, and dataset on TF Hub. Also, we aim to broaden the scope of MIEV's capabilities including not only ML developers but also non-AI experts and policymakers involved in regulating AI to support carbon-conscious decision-making. Additionally, we intend to assess the usability of MIEV and enhance its functionality. Thus, we plan to employ target users testing methodologies and gather feedback from target users to identify areas of improvement in the tool's usability and design. Furthermore, we aim to evaluate the efficacy of MIEV in potentially reducing carbon intensity. We aim to conduct a user study to assess MIEV's impact on users' understanding of the carbon intensity that ML models produce during the **Inference** phase.

7 CONCLUSION

The goal of this study was to help ML developers consider carbon intensity when developing models for sustainable AI. The primary contribution of this study is to develop MIEV, an interactive tool visualizing the carbon intensity for ML models when inferring data. MIEV enables ML developers to view the carbon intensity of image classification models on TF Hub. Also, our proposed tool allows users to compare the carbon intensity produced at each inference step with classification results when models infer data uploaded by the users. Future work remains to expand our tool's capability to include models from different domains, such as audio and video on TF Hub, as well as evaluate its usability and feasibility with multiple stakeholders involving model publishers, and online community operators. We hope our work inspires researchers to create technologies for facilitating online communities of ML developers towards sustainable AI.

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